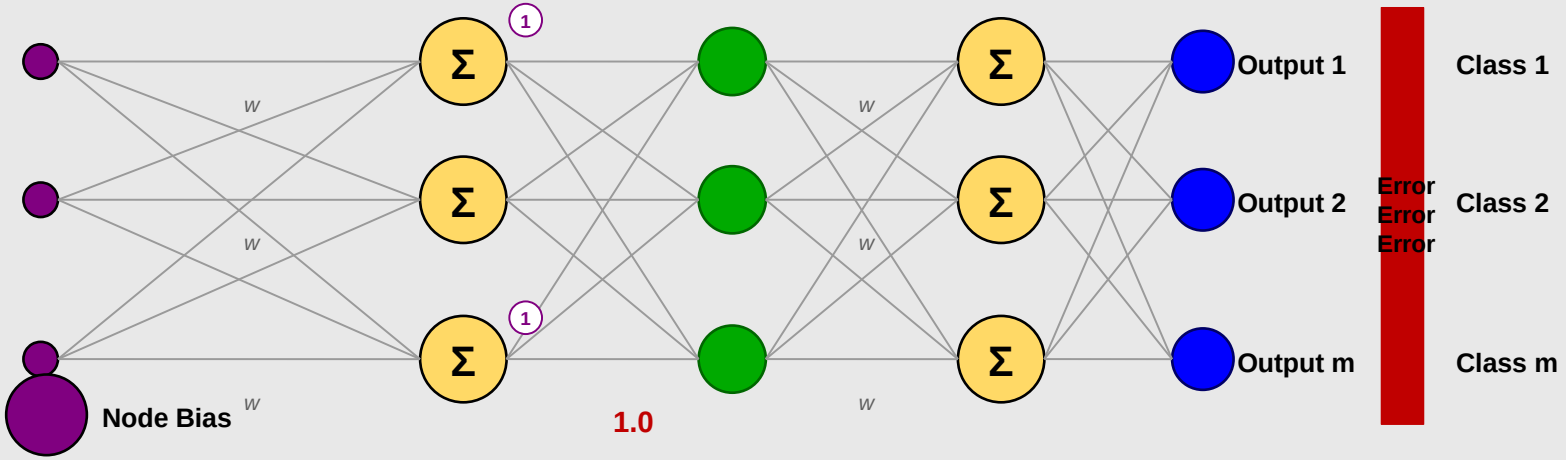
Vanishing and Exploding Gradients

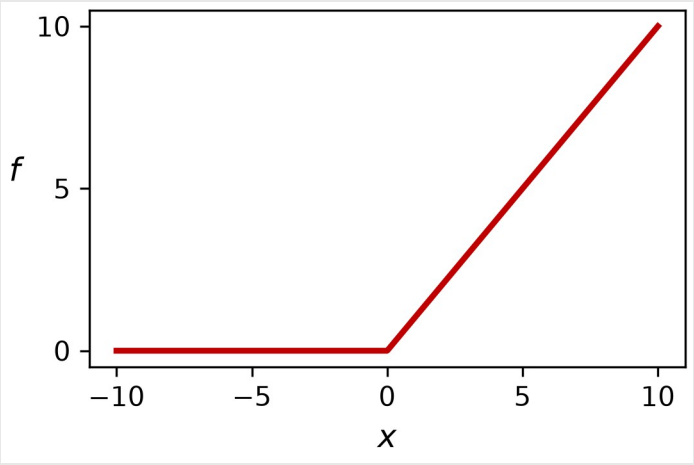
Long Chain of Derivatives

$$f(x) = \text{ReLU} = \begin{cases} 0 \text{ for } x < 0 \\ x \text{ for } x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} 0 \text{ for } x < 0 \\ 1 \text{ for } x \geq 0 \end{cases}$$



Exploding Gradients



Vanishing Gradients

